Year 9 · Chapter 13 · Similar figures: answers (teacher copy)

13.1 Similar figures (booklet p. 3)

WE DO examples (teacher)

1. Ex 1a: FE and TR, EC and RM, CD and MP, DF and PT
2. Ex 1b: ∠F and ∠T, ∠E and ∠R, ∠C and ∠M, ∠D and ∠P
3. Ex 1c: FECD ||| TRMP
4. Ex 1d: CD matches MP; Scale factor = 6 ÷ 4 = 3/2
5. Ex 2a: AB and YZ (6 and 4), BC and ZX (9 and 6), CA and XY (12 and 8)
6. Ex 2b: ABC ||| YZX
7. Ex 2c: Scale factor = 4 ÷ 6 = 2/3: a reduction (less than 1)
8. Ex 2d: ∠B matches ∠Z

Easy

1a AB matches PQ; Scale factor = 6 ÷ 3 = 2

1b AB matches PQ; Scale factor = 8 ÷ 4 = 2

2a CA matches LM; Scale factor = 2.5 ÷ 5 = 1/2

2b AB matches FD; Scale factor = 6 ÷ 4 = 3/2

3a AB matches PQ (3 and 6 squares); Scale factor = 6 ÷ 3 = 2

3b BC matches SP (4 and 2 squares); Scale factor = 2 ÷ 4 = 1/2

4a ABC ||| YZX; Scale factor = 6 ÷ 4 = 3/2

4b ABC ||| NLM; Scale factor = 3 ÷ 6 = 1/2

Medium

5 ABC ||| RPQ

6 12 : 18 = 2 : 3 (Yr 8)

7 Scale factor = 7.5 ÷ 5 = 3/2

8 QR matches YZ; ∠P matches ∠X

9 P(less than 3) = 2/6 = 1/3 (12.1)

10 Scale factor = 4 ÷ 6 = 2/3: a reduction

Worded problems

1. AB and KL, BC and LM, CA and MK: ABC ||| KLM; Scale factor = 9 ÷ 6 = 3/2 (also 12 ÷ 8 = 3/2)
2. 4.2 m = 420 cm; Scale factor = 420 ÷ 12 = 35
3. All squares are similar: every angle is 90° and every side is enlarged by the same scale factor. Not all rectangles: 2 ÷ 1 = 2 but 3 ÷ 2 = 3/2, so the sides are not in the same ratio

13.2 Enlargements (booklet p. 6)

WE DO examples (teacher)

1. Ex 1a: 2 by 1; image 4 by 2 squares
2. Ex 1b: base 2, height 1; image base 6, height 3 squares
3. Ex 1c: 4 by 2; image 2 by 1 squares
4. Ex 1d: 2, 1, 1, 1, 1, 2; image 4, 2, 2, 2, 2, 4 squares (going round from the bottom left)
5. Ex 2a: 4 × 3 = 12 cm, 6 × 3 = 18 cm
6. Ex 2b: 4 × 1/2 = 2 cm, 6 × 1/2 = 3 cm
7. Ex 2c: 4 × 2.5 = 10 cm, 6 × 2.5 = 15 cm
8. Ex 2d: Scale factor = 6 ÷ 4 = 3/2 (also 9 ÷ 6 = 3/2)

Easy

1a Original lengths: 3 by 2; Image lengths: 6 by 4 (squares)

1b Original lengths: 2 by 1; Image lengths: 6 by 3 (squares)

2a Original lengths: base 3, height 2; Image lengths: base 6, height 4 (squares)

2b Original lengths: base 6, height 4; Image lengths: base 3, height 2 (squares)

3a Original lengths: 2, 1, 1, 1, 1, 2; Image lengths: 6, 3, 3, 3, 3, 6 (squares)

3b Original lengths: 4, 2, 2, 2, 2, 4; Image lengths: 2, 1, 1, 1, 1, 2 (squares)

4a Image lengths: 3, 2, slanted 1 across and 1 up, 1, 1, 1, 2 (squares, going round from the bottom left)

4b Image lengths: 2, 1, 1, 2, 2, slanted 1 across and 1 down, 2 (squares, going round from the bottom left)

Medium

5 6 by 3 squares

6 3 + 5 = 8 parts; $40 ÷ 8 = $5; $15 and $25 (Yr 8)

7 10 cm by 15 cm

8 Scale factor = 2 ÷ 4 = 1/2

9 HH, HT, TH, TT: P(two heads) = 1/4 (12.4)

10 1.5 cm, 2 cm, 2.5 cm

Worded problems

1. 12 × 3 = 36 cm, 8 × 3 = 24 cm: 36 cm by 24 cm
2. 4.2 m = 420 cm; 420 × 1/20 = 21 cm
3. Original area 3 × 3 = 9 cm²; image 6 × 6 = 36 cm²; 36 ÷ 9 = 4 times

13.3 Finding unknown sides (booklet p. 9)

WE DO examples (teacher)

1. Ex 1a: Scale factor = 6 ÷ 3 = 2; x = 4 × 2 = 8 cm
2. Ex 1b: Scale factor = 4 ÷ 2 = 2; x = 4 × 2 = 8 cm
3. Ex 1c: Scale factor = 10 ÷ 5 = 2; x = 8 ÷ 2 = 4 cm
4. Ex 1d: Scale factor = 6 ÷ 4 = 3/2; x = 5 × 3/2 = 7.5 cm
5. Ex 2a: 29 ÷ 20 = 1.45, 18 ÷ 12 = 1.5: not all equal, so not similar
6. Ex 2b: In order of size: 5.25 ÷ 7 = 0.75, 6 ÷ 8 = 0.75, 7.5 ÷ 10 = 0.75: all equal, so similar (scale factor 6 ÷ 8 = 3/4)
7. Ex 2c: Scale factor = 6 ÷ 8 = 3/4; EF = 12 × 3/4 = 9 cm
8. Ex 2d: AC = 7.5 ÷ 3/4 = 10 cm

Easy

1a Scale factor = 6 ÷ 3 = 2; x = 5 × 2 = 10 cm

1b Scale factor = 6 ÷ 2 = 3; x = 5 × 3 = 15 cm

2a Scale factor = 6 ÷ 4 = 3/2; x = 6 × 3/2 = 9 cm

2b Scale factor = 16 ÷ 8 = 2; x = 12 ÷ 2 = 6 cm

3a Scale factor = 5 ÷ 10 = 1/2; x = 8 × 1/2 = 4 cm

3b Scale factor = 5 ÷ 2 = 5/2; x = 3 × 5/2 = 7.5 cm

4a Scale factor = 9 ÷ 6 = 3/2; x = 4 × 3/2 = 6 cm

4b Scale factor = 3.5 ÷ 7 = 1/2; x = 4 ÷ 1/2 = 8 cm

Medium

5 9 ÷ 6 = 1.5, 6 ÷ 4 = 1.5: all equal, so similar

6 45 : 60 = 3 : 4 (Yr 8)

7 Scale factor = 10 ÷ 5 = 2; x = 4 × 2 = 8 cm

8 6 ÷ 3 = 2, 8 ÷ 4 = 2, 9 ÷ 5 = 1.8: not all equal, so not similar

9 Observed P(head) = 28/50 = 14/25 (12.2)

10 Scale factor = 10 ÷ 4 = 5/2; EF = 6 × 5/2 = 15 cm

Worded problems

1. Scale factor = 20 ÷ 8 = 5/2; width = 5 × 5/2 = 12.5 cm
2. Scale factor = 25 ÷ 10 = 5/2; 6 × 5/2 = 15 cm, 8 × 5/2 = 20 cm
3. 16 ÷ 80 = 1/5 and 9 ÷ 45 = 1/5: the same scale factor, so the screens are similar

13.4 Shadows and scale drawings

(booklet p. 12)

WE DO examples (teacher)

1. Ex 1a: Scale factor = 14 ÷ 2 = 7; h = 1 × 7 = 7 m
2. Ex 1b: Scale factor = 6 ÷ 1.2 = 5; h = 1.8 × 5 = 9 m
3. Ex 1c: Scale factor = 100; AB = 5.5 × 100 = 550 cm = 5.5 m
4. Ex 1d: Scale factor = 25 000; d = 8 × 25 000 = 200 000 cm = 2 000 m = 2 km
5. Ex 2a: AB = 6 × 200 = 1 200 cm = 12 m
6. Ex 2b: CD = 4 × 200 = 800 cm = 8 m
7. Ex 2c: 9 m = 900 cm; 900 ÷ 200 = 4.5 cm
8. Ex 2d: 0.8 m = 80 cm; 80 ÷ 200 = 0.4 cm

Easy

1a 8 m = 800 cm; Scale factor = 800 ÷ 80 = 10; h = 50 × 10 = 500 cm = 5 m

1b Scale factor = 12 ÷ 1.5 = 8; h = 1 × 8 = 8 m

2a Scale factor = 10 ÷ 2 = 5; h = 1.5 × 5 = 7.5 m

2b 2 m = 200 cm; 9 m = 900 cm; Scale factor = 900 ÷ 200 = 9/2; h = 160 × 9/2 = 720 cm = 7.2 m

3a Scale factor = 200; AB = 4 × 200 = 800 cm = 8 m

3b Scale factor = 5 000; d = 6 × 5 000 = 30 000 cm = 300 m

4a Scale factor = 24 ÷ 3 = 8; h = 2 × 8 = 16 m

4b 1.2 m = 120 cm; 6 m = 600 cm; Scale factor = 600 ÷ 80 = 15/2; h = 120 × 15/2 = 900 cm = 9 m

Medium

5 Scale factor = 8 ÷ 1.6 = 5; h = 1 × 5 = 5 m

6 1 + 4 = 5 parts; 60 ÷ 5 = 12; 12 and 48 (Yr 8)

7 d = 3 × 50 000 = 150 000 cm = 1 500 m = 1.5 km

8 Scale factor = 50; AB = 7 × 50 = 350 cm = 3.5 m

9 HH, HT, TH, TT: P(at least one head) = 3/4 (12.4)

10 9 m = 900 cm; 900 ÷ 200 = 4.5 cm

Worded problems

1. 20 m = 2 000 cm; Scale factor = 2 000 ÷ 150 = 40/3; h = 60 × 40/3 = 800 cm = 8 m
2. Real length = 35 × 48 = 1 680 cm = 16.8 m
3. d = 18 × 25 000 = 450 000 cm = 4 500 m = 4.5 km